CST2550 - Coursework 2

Brainstorming

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1. Introduction

This document presents the proposed ideas for the project, including both implemented and unimplemented features. These are discussed in detail throughout the report.

Figure 1 illustrates the ideas gathered for the application, primarily during team meetings, covering features for both user and administrator roles, as well as the implementation of the AI Agent.

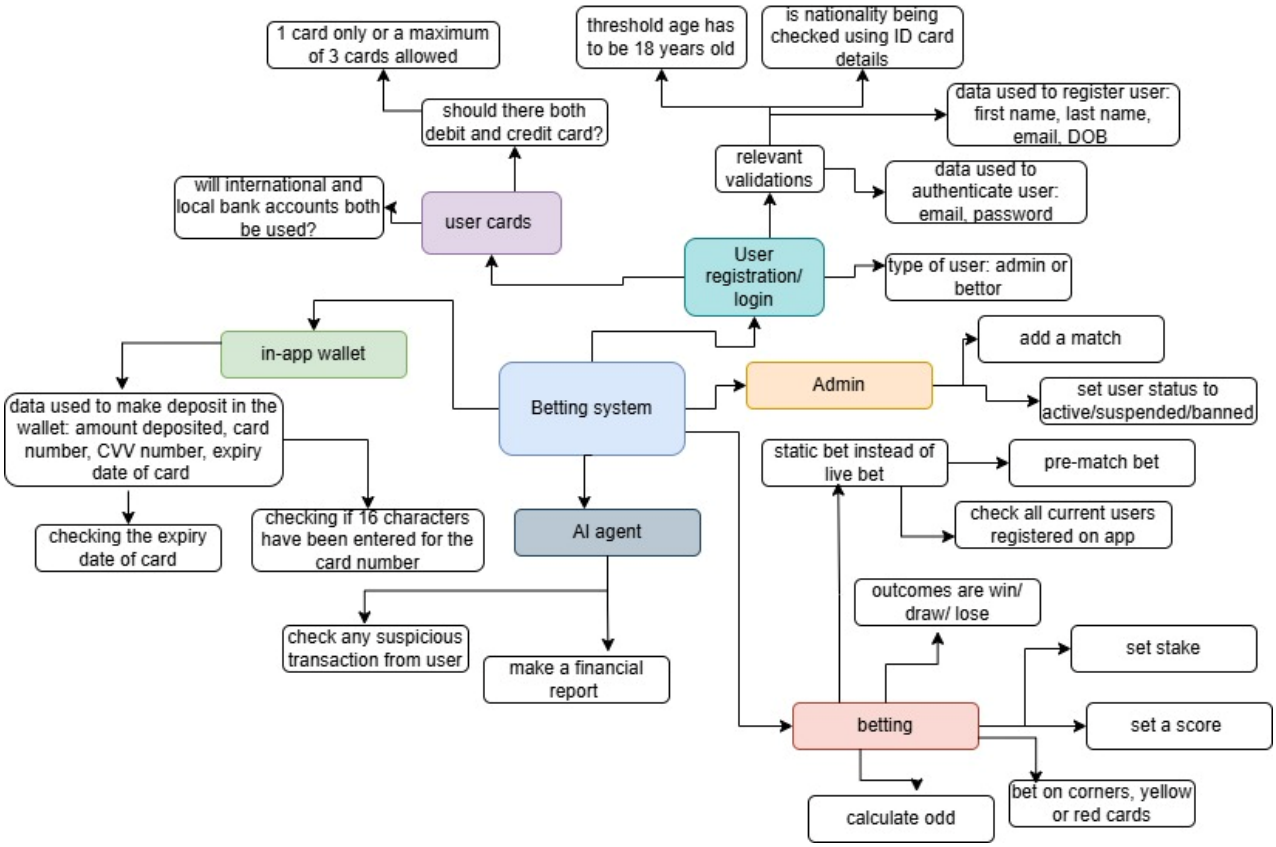

Figure 1: Brainstorm of General Ideas Gathered for the System

2. Proposed Ideas

2.1 Type of Bets

To reflect common football betting options, the system should support at least the following types of bets:

1. Basic Match Result Bets
   - **1X2 (Match Outcome):** A bet on whether the home team wins (1), the match is a draw (X), or the away team wins (2).
   - **Double Chance**: A safer bet covering two outcomes (e.g., Home Win OR Draw).
   - **Correct Score**: A high-odds bet on the exact final score (e.g., 2-1 or 0-0).

2. Goal-Based Bets
   - **Over/Under Goals**: Betting if the total goals will be over or under a specific number, usually 2.5.
   - **Both Teams to Score (BTTS):** A simple Yes/No data field that checks if both teams found the net.
   - **First Goalscorer**: A bet on a specific player.
3. Statistical Bets
   - **Total Corners**
   - **Number of Yellow and Red Cards**

2.2 Implemented Features

- Filtering and Refreshing
  - A filtering feature to allow users to view games from selected leagues on both both administrator and user interface.
  - A search feature to enable users to find football matches based on the team names.
  - A filtering feature in the bet history section to sort bet slips by status (won/lost/pending) and by date (newest to oldest).
  - A refresh feature to reload data from the database as most data will be fetched only once initially.

- Notification
  - Displaying a message to inform users of successful operations, errors, or warnings.

- Updates
  - An automatic update mechanism to simulate live matches.
  - Remove completed and started matches from the main page on user's side.
  - Update status of match and bets accordingly.

2.3 Unimplemented Features

- Handicap Bet
  - A handicap betting system that gives one team an advantage before the match begins.
  - For example, if a before selects a handicap and the chosen team is given a +1 goal advantage, a match ending 0-0 would be adjusted to 1-0
  - This allows bets to still be considered wins even if the actual match result is a draw, depending on the handicap applied.

- This type of bet would have provided a safer betting option when selecting weaker teams.
- It would have been a separate bet type just like match outcomes, red/yellow cards and goal scorers.
- However, it was not implemented due to the complexity of the logic required to adjust scores and the need for additional data for accurate calculations.

➢ Detection of Suspicious Users
  - This feature would have been implemented by training a machine learning model using a dataset containing features to identify suspicious user activity (e.g., betting patterns and failed login attempts)
  - User activity would be recorded in the database, and AI agent would analyse this data to detect suspicious behaviour and flag or suspend them.
  - This feature was not implemented due to time constraints and the lack of suitable dataset for training.

➢ User Authentication using Official Systems
  - Having this criterion would imply authenticating the user using an NIC or passport number from their home country, therefore arising the need to validate the user's identity from their birth country.
  - Similarly, this case would imply that the user's bank account will have to be checked whether it is a local or international bank account and how many accounts are active under that user's identity.
  - Therefore, this criterion for validation has been omitted since a big amount of validation will have to be made by comparing large amounts of credible data for passport number or NIC.
  - Since this is sensitive data collection, a privacy policy would then have to be agreed upon by the end-user before accessing the betting application.

2.4 Admin Side

- A feature to add matches with validation rules to ensure that new matches start at least 5 minutes after the current time and that the home and away teams are not the same.
- A match management interface to display all matches and allow browsing, with filtering options by leagues, and search functionality by team name.
- A feature to display match outcomes of completed matches.
- A system update mechanism where match status is updated automatically through the simulation feature.

2.5 AI Agent

2.5.1 Core Concept

The AI agent is based on a reasoning model capable of understanding text and making decisions, with intelligent responses and automated analysis.

2.5.2 Potential AI Agent Features

- Flagging suspicious accounts based on activity patterns.
- Predicting odds.
- Providing betting advice to users.
- Generating financial reports to analyse revenue and system performance
- Navigation assistant and customer support through chatbot

3. Mock Interface Design

Before developing the application's frontend, the interface design was drafted using Canva for team approval and to guide implementation. The full design can be accessed via https://www.canva.com/design/DAHAjsXvIhI/ASuEQJcFaCiJfUWaguSrYA/edit. The figures below illustrate the different interface mock-ups created.

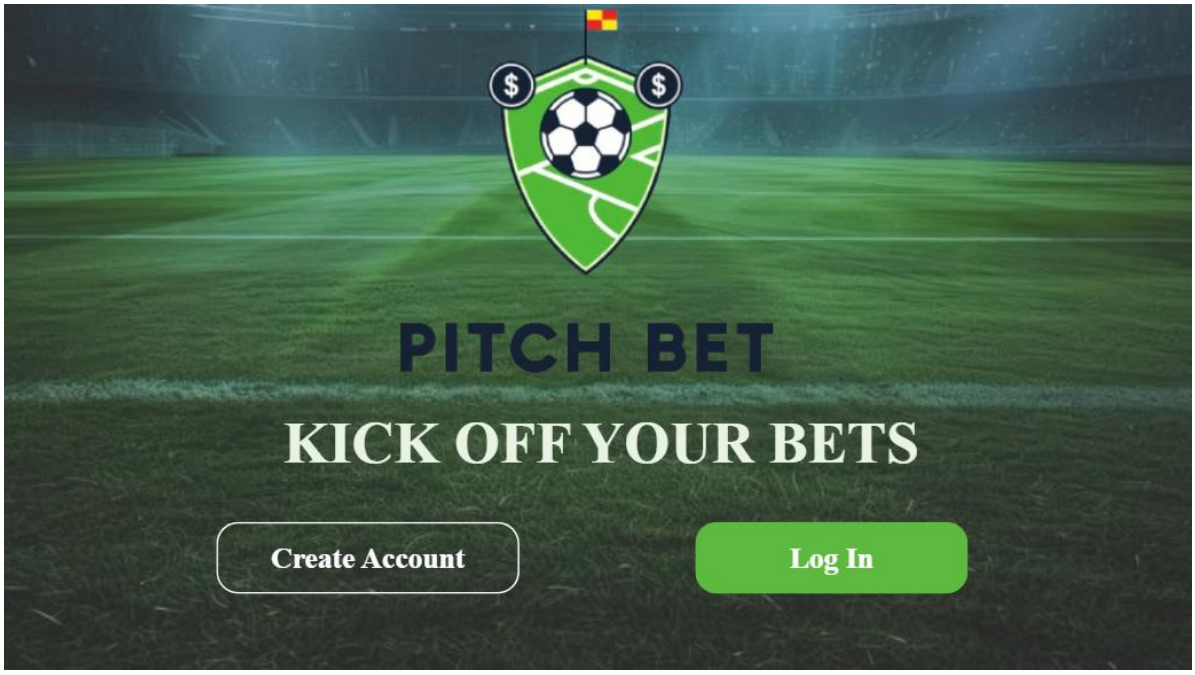

Figure 2: Draft Landing Page Design

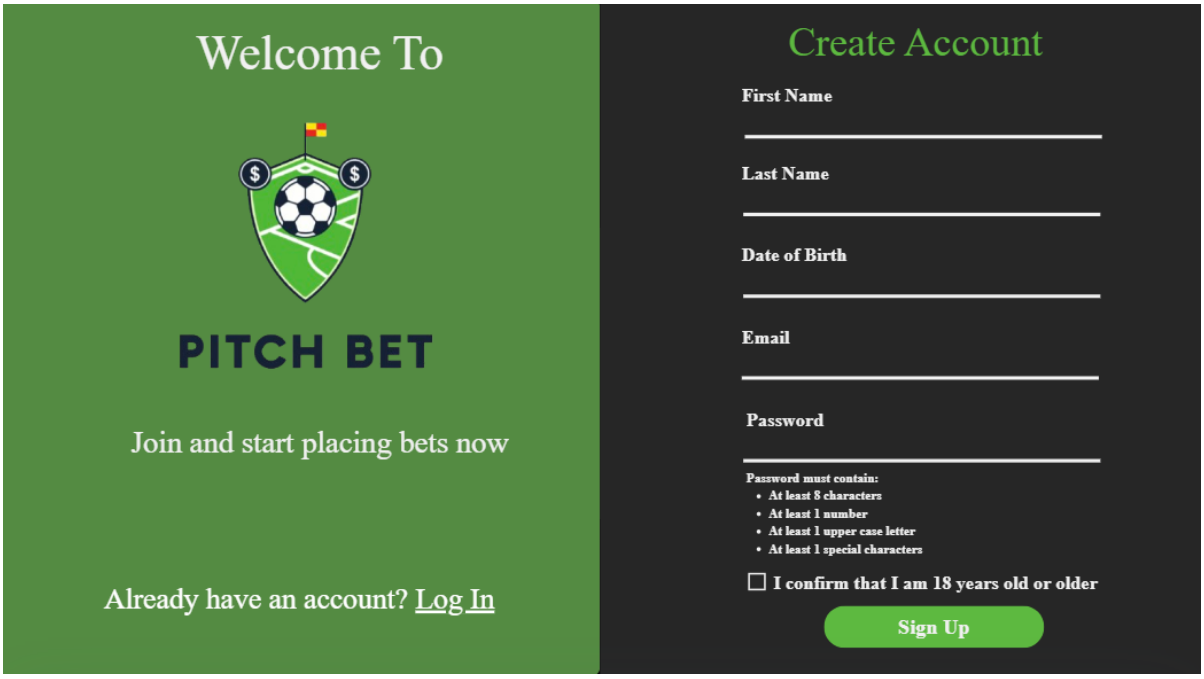

Figure 3: Draft Registration Page Design

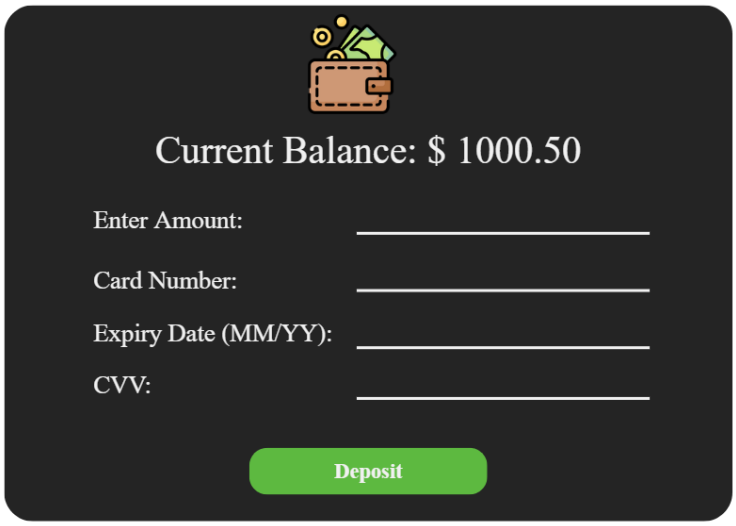

Figure 4: Draft Deposit Popup Design

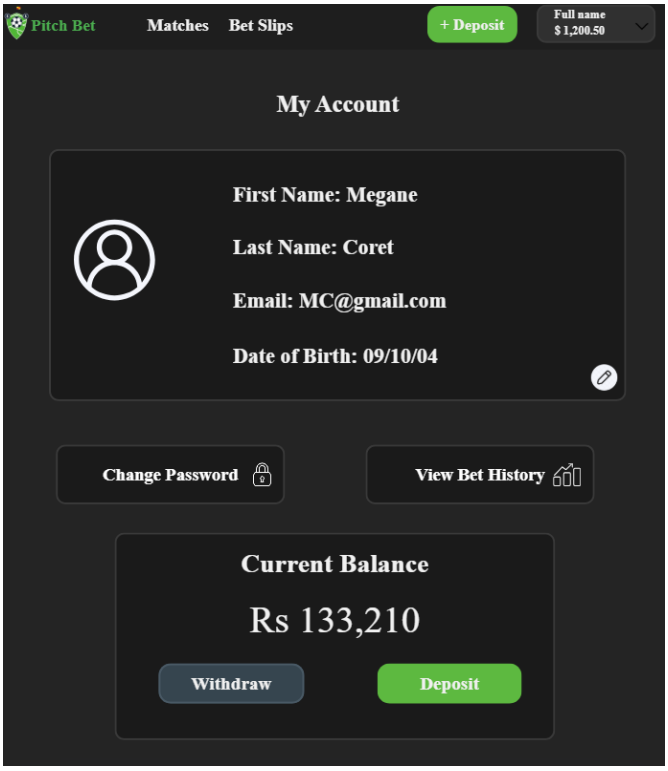

Figure 5: Draft Profile Page Deposit

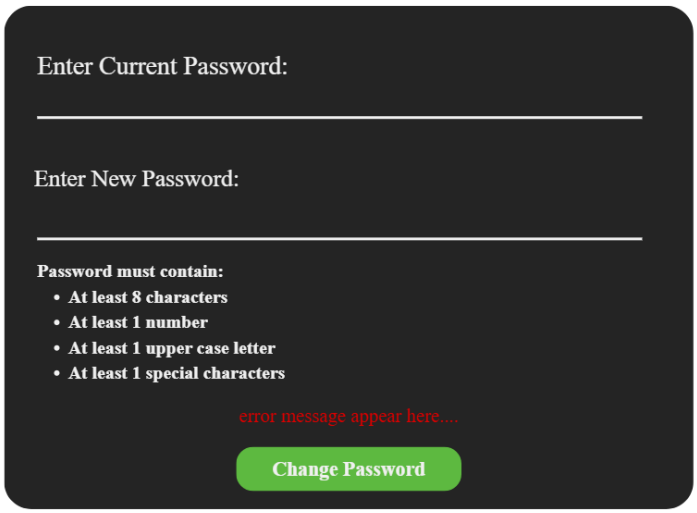

Figure 6: Draft Password Change Popup Deposit

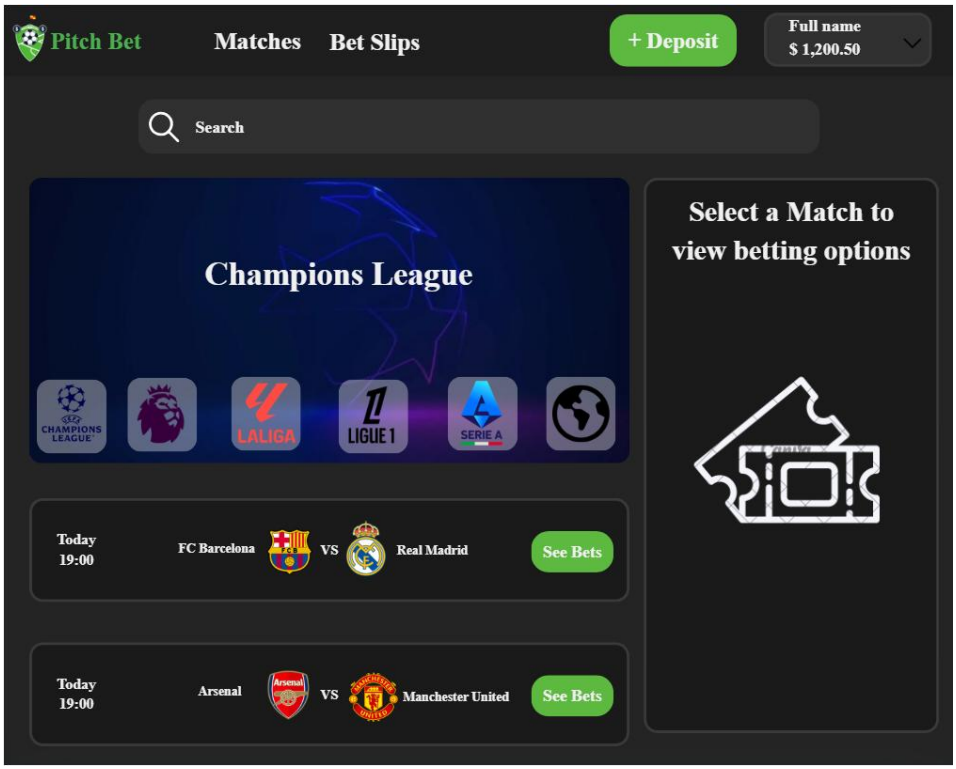

Figure 7: Draft Main Page Design

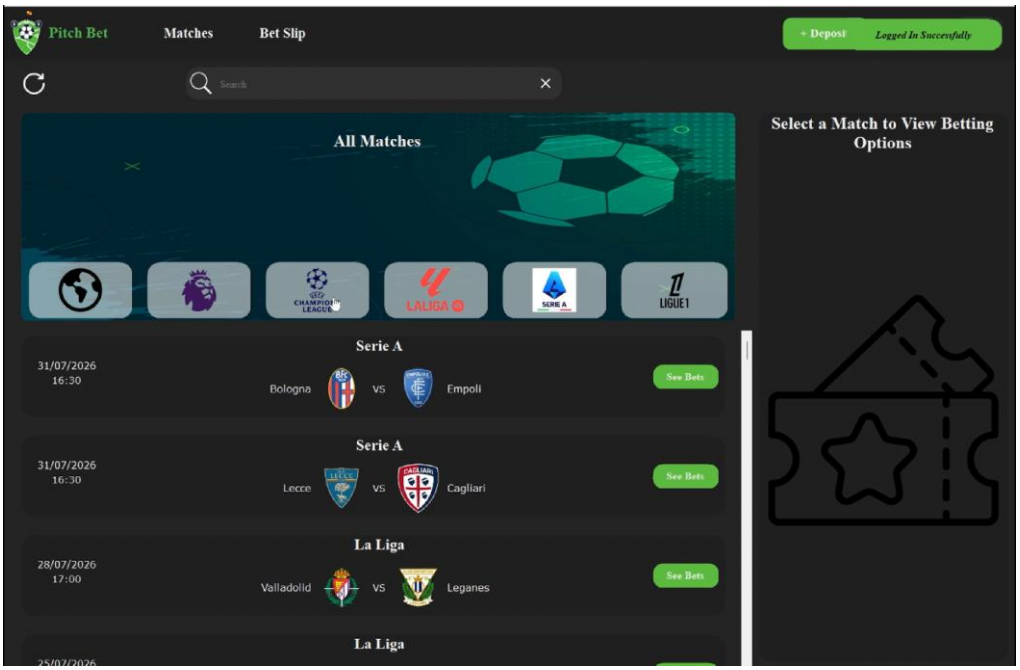

Figure 8: Main Page Implementation (WinForms)

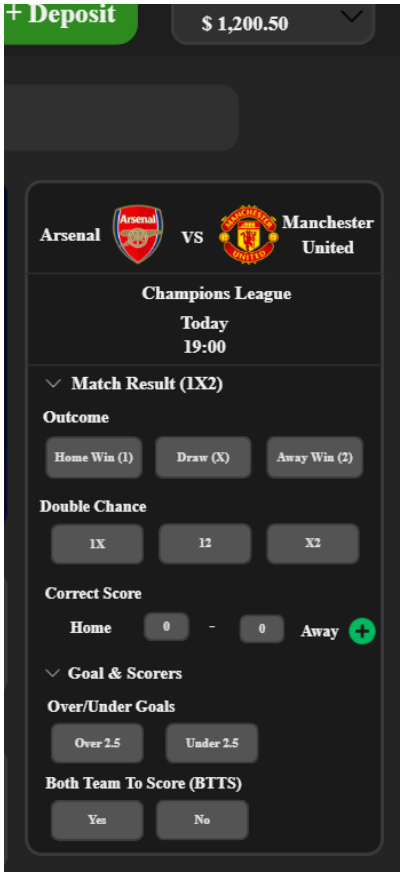

Figure 9: Draft Bet Panel Design

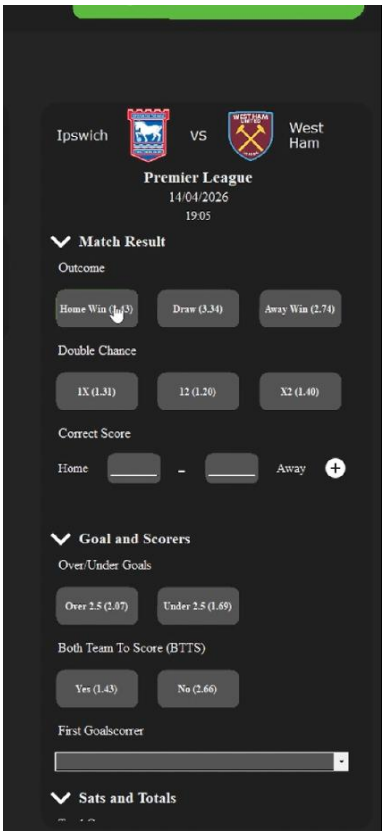

Figure 10: Bet Panel Implementations (WinForms)

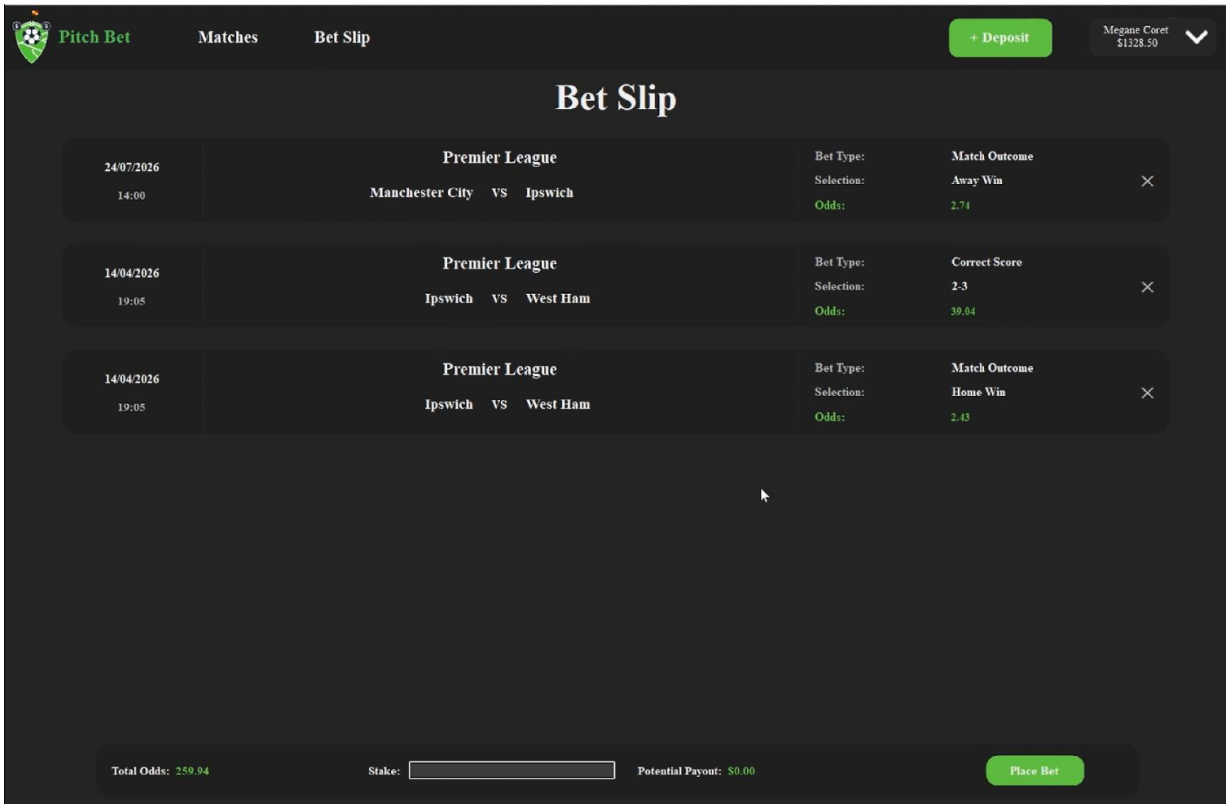

Figure 11: Bet Slip Page Implementation (WinForms)

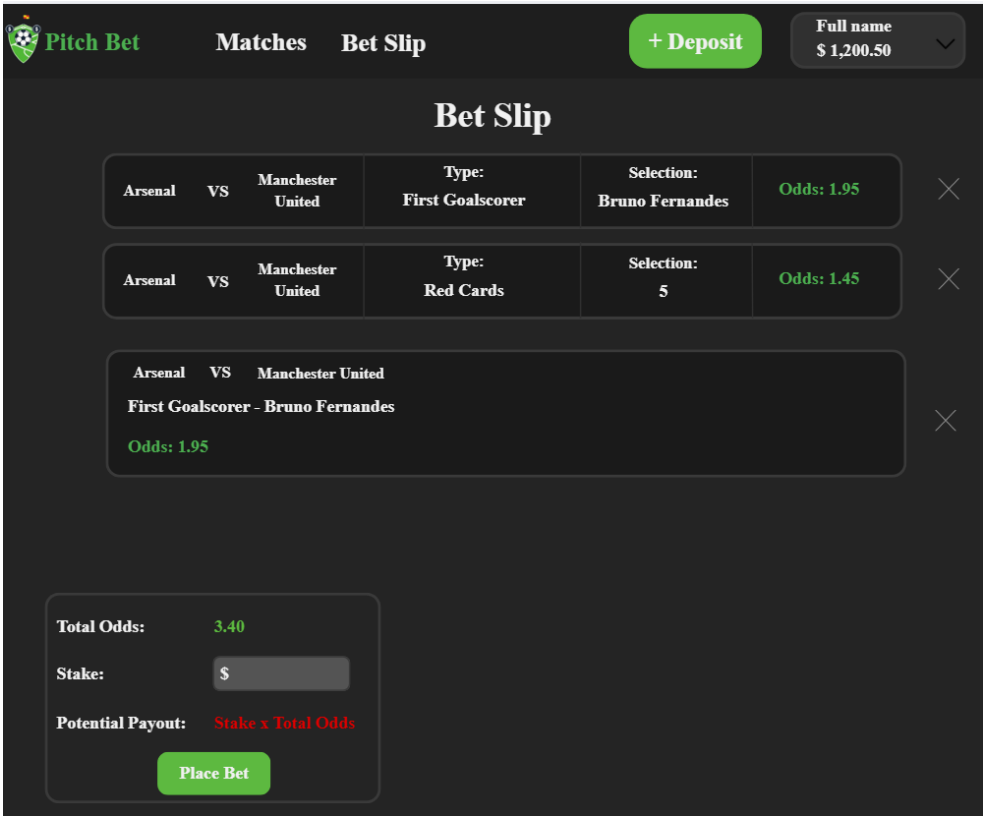

Figure 12: Draft Bet Slip Page Design

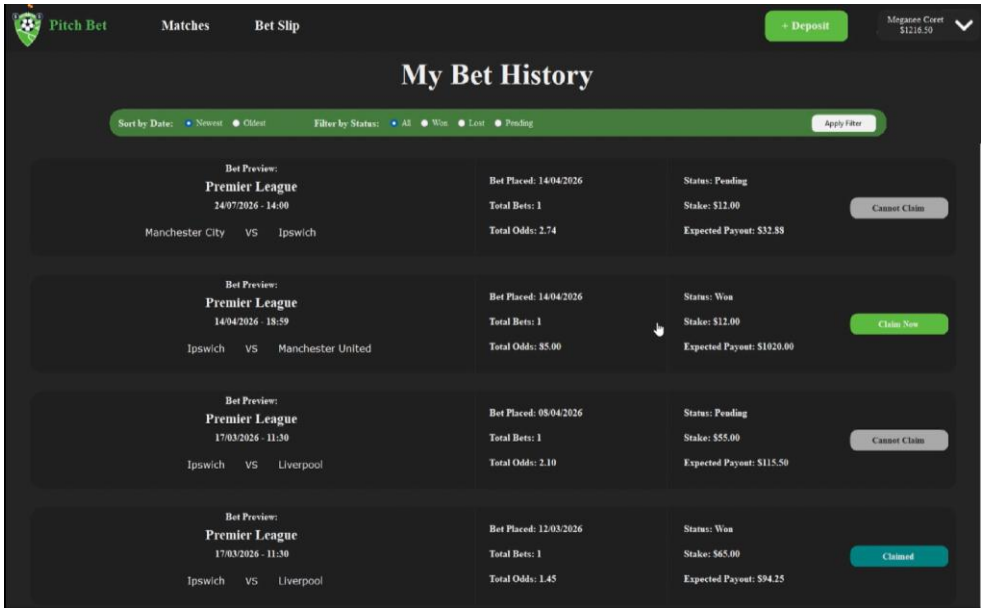

Figure 13: Bet History Page Implementation (WinForms)

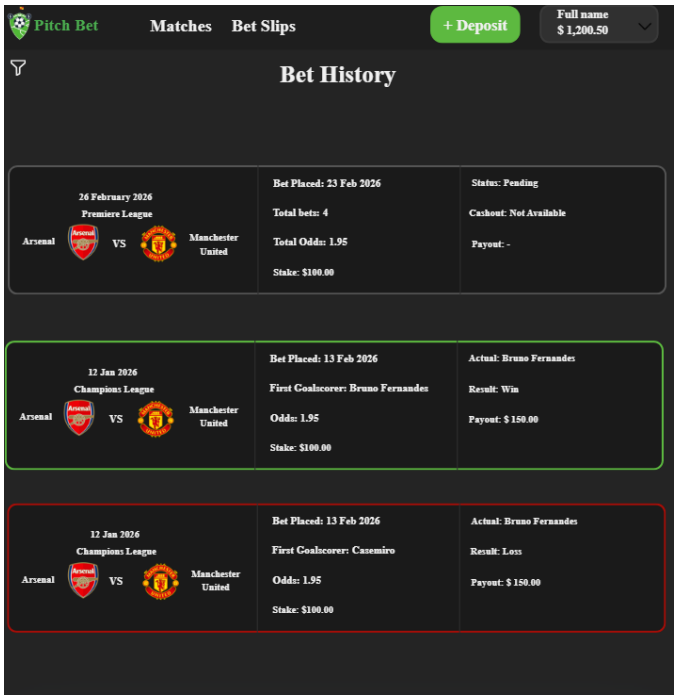

Figure 14: Draft Bet History Page Design (1)

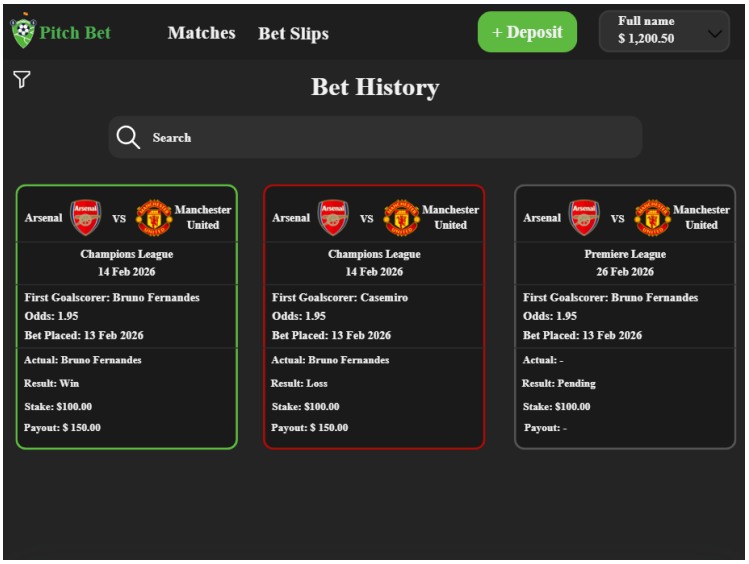

Figure15 15: Draft for Bet History Page Design (2)